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De Finetti-type theorems for nonexchangeable 0–1 random variables

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Abstract We impose conditions on a family of 0–1 random variables ensuring that they verify a de Finetti-type theorem. More precisely, we assume that the 0–1 random variables are the indicator functions of a random selection process and, under fairly weak hypotheses (which, in particular, generalize the previously existing results), we prove that they verify a de Finetti theorem, though they might neither be exchangeable nor identically distributed.

Keywords De Finetti theorem · Exchangeability · Random selection processes

Mathematics Subject Classification (2000) 60G09

1 Introduction

The classical de Finetti theorem establishes that an infinite sequence of exchangeable random variables is conditionally independent and identically distributed. This result was shown, firstly, for $\{0, 1\}$-valued random variables (see, e.g., Feller 1966, Chap. VII, §4). Subsequently, the de Finetti theorem was extended to the case of exchangeable sequences of random variables taking values in more general spaces; see, for instance, Chow and Teicher (1988, Chap. 7.3), Fristedt and Gray (1997, Chap. 27), Hewitt and Savage (1955), or Kallenberg (1997, Chap. 8). The reader interested in exchangeability issues can also consult (Aldous 1985; Kallenberg 2005).

Our interest is to prove a de Finetti-type theorem for a class of $\{0, 1\}$-valued random variables that are not identically distributed. More precisely, given an infinite set

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E and a family of random indicator functions

$$N(x) \in \{0, 1\}, \quad \text{for } x \in E,$$

we say that the random variables $N(x)$ verify a de Finetti-type theorem if they are conditionally independent given some random variable. The non-identically distributed case for such de Finetti-type theorems is related to the construction of the Martin boundary associated to a generalized Stirling triangle of the first kind; see (Gnedin and Pitman 2006) or (Vélez Ibarrola and Prieto-Rumeau 2009, Sect. 4) for a brief summary. Usually, such results establish that the set of distributions of $\{N(x)\}_{x\in E}$ is a convex set, whose extreme distributions correspond to the case of independence of the $N(x)$. The whole family of distributions is then obtained by integration with respect to a mixing probability distribution over some parameter that identifies the extremes (see Diaconis and Freedman 2004); so that, conditional on this parameter, the random variables $N(x)$ are independent.

Our experience, however, is that an adequate setting to prove extensions of de Finetti's theorem is by means of the so-called *random selection processes*; see (Vélez Ibarrola and Prieto-Rumeau 2009) for the case of $\{0, 1\}$-valued random variables, and (Vélez Ibarrola and Prieto-Rumeau 2008) for finite-valued random variables.

The definition of a random selection process, introduced in Vélez Ibarrola and Prieto-Rumeau (2009), is the following. Suppose that we are given an arbitrary infinite set E and a *weight function* $m : E \to (0, \infty)$. A random selection process $\boldsymbol{N}$ on (E, m) is a family of random variables

$$N(x) \in \{0, 1\}, \quad \text{for } x \in E,$$

defined on a common probability space $(\Omega, \mathcal{F}, \mathrm{P})$, that verify, for every finite subset $A \subset E$, the following conditions:

C1. The random variables $\{N(x)\}_{x\in A}$ and $\{N(x)\}_{x\in A^c}$ are conditionally independent given $N(A)$;

C2. $\mathrm{P}\{N(x) = 1 \mid N(A) = 1\} = m(x)/m(A)$ for all $x \in A$,

where $N(A) := \sum_{x\in A} N(x)$ and $m(A) := \sum_{x\in A} m(x)$. The random selection process $\boldsymbol{N}$ will refer to $\{N(x)\}_{x\in E}$ as well as to $\{N(B)\}_{B\in\mathcal{B}}$, where $\mathcal{B}$ is the ring of finite subsets of E.

We can interpret that $N(x) = 1$ means that the element $x \in E$ is "selected", while $x \in E$ is "rejected" when $N(x) = 0$. The Markovian Condition **C1** means that, once the number of selected points in A is known, the corresponding choices inside A do not depend on the selections made outside A. Condition **C2** establishes that, if a single point in A is selected, then each point $x \in A$ has probability proportional to $m(x)$ of being selected.

It is worth mentioning that exchangeability is not part of our framework here. Indeed, in this paper, we base our approach on Conditions **C1** and **C2**. Nevertheless, to compare the above conditions to exchangeability, we state the following preliminary proposition, whose proof is given in Sect. 4.

Proposition 1.1 *A family of exchangeable indicator random variables $\{N(x)\}_{x\in E}$ is a random selection process with constant weight function.*

Conversely, if $\boldsymbol{N}$ *is a random selection process on* (E, m) *with constant weight function, then the random variables* $\{N(x)\}_{x\in E}$ *are exchangeable.*

The above proposition shows that the exchangeability condition is equivalent to the Markovian Condition **C1** and, in addition, the conditionally uniform selection of points within any finite set (Condition **C2**).

In this paper, we give general conditions on the weight function m of a random selection process $\boldsymbol{N}$ on (E, m) ensuring that the indicator functions $\{N(x)\}_{x\in E}$ are conditionally independent given a certain random variable. In Vélez Ibarrola and Prieto-Rumeau (2009), two such de Finetti-type theorems have been proved for an arbitrary infinite set E when the weight function contains a constant or a linear sequence; that is, when there exists a denumerable set $\{x_i^\star\}_{i\geq 1} \subseteq E$ and some $h > 0$ such that either

$$m\big(x_i^\star\big) = h \quad \text{or} \quad m\big(x_i^\star\big) = ih \quad \text{for all } i \geq 1. \tag{1.1}$$

The proof of the former result uses the exchangeability of the sequence $\{N(x_i^\star)\}_{i\geq 1}$, while the proof of the latter is an application of the Hausdorff moment problem technique.

In this paper, new methods are used to derive the following result: If there exists a denumerable set $\{x_i^\star\}_{i\geq 1} \subseteq E$ such that either

- $\{m(x_i^\star)\}$ is nonincreasing and $\sum m(x_i^\star) = \infty$, or
- $\{m(x_i^\star)\}$ is nondecreasing and $\sum 1/m(x_i^\star) = \infty$,

then the random variables $\{N(x)\}_{x\in E}$ are conditionally independent. Obviously, our results here generalize the cases in (1.1); see (Vélez Ibarrola and Prieto-Rumeau 2009). It is worth mentioning that it suffices that there exists a denumerable set $\{x_i^\star\}_{i\geq 1} \subseteq E$ satisfying the above conditions to deduce that the de Finetti-type theorem holds for *all* the random variables $\{N(x)\}_{x\in E}$.

The rest of the paper is organized as follows. In Sect. 2, we recall some useful results on random selection processes and we state our main theorems which, in particular, include a strong law of large numbers. An example of a random selection process is proposed in Sect. 3. Section 4 contains the proofs of the results in Sect. 2. In Sect. 5, the properties of the so-called rejection function of the random selection process are analyzed. Some final comments are included in Sect. 6.

2 Model definition and main results

We recall some results on the conditional and marginal distributions of a random selection process. For details, the reader is referred to Vélez Ibarrola and Prieto-Rumeau (2009).

Marginal and conditional distributions. First of all, let us introduce some notation. Given a finite subset A of E and $0 \leq r \leq \#A$ (in the forthcoming, $\#A$ denotes the cardinal of the set A), we define $M_r(A)$ as the sum of the products $\prod_{j=1}^r m(x_j)$ over

all the subsets $\{x_1, \ldots, x_r\}$ of A with r elements. Equivalently, $M_r(A)$ is the rth degree coefficient of the polynomial

$$\prod_{x \in A} [1 + m(x)z]$$

(note that if $m(x) = 1$ for every $x \in A$, then $M_r(A)$ equals the binomial coefficient $\binom{\#A}{r}$). To simplify the notation, let $M_r(A) := 0$ whenever $r < 0$ or $r > \#A$.

It is proved in Vélez Ibarrola and Prieto-Rumeau (2009, Proposition 2.3) that the conditional distributions of a random selection process $\boldsymbol{N}$ are given by

$$\mathrm{P}\big\{N(B) = s \mid N(A) = r\big\} = \frac{M_s(B) M_{r-s}(A - B)}{M_r(A)} \tag{2.1}$$

for finite sets $B \subseteq A \subseteq E$, and $0 \le r \le \#A$. Hence, the conditional distributions of a random selection process are *necessarily* of this "hypergeometric type". Besides, to determine the distribution of $\boldsymbol{N}$, we also need a family of marginal distributions $\mathrm{P}\{N(B) = s\}$ that is compatible with (2.1), meaning that the "total probability rule"

$$\mathrm{P}\big\{N(B) = s\big\} = \sum_{r=0}^{\#C} \mathrm{P}\big\{N(B \cup C) = s + r\big\} \frac{M_s(B) M_r(C)}{M_{s+r}(B \cup C)} \tag{2.2}$$

holds for every finite disjoint sets B and C, and every $0 \le s \le \#B$.

Note that there exist two obvious degenerate (non-random) solutions: $N(B) \equiv 0$ and $N(B) \equiv \#B$. Moreover, it is proved in Vélez Ibarrola and Prieto-Rumeau (2009) that marginal distributions of the "binomial type" satisfy (2.2). More precisely, given an arbitrary probability distribution F on $[0, 1]$, the marginal distributions

$$\mathrm{P}\big\{N(B) = s\big\} = \int_0^1 \frac{M_s(B) \xi^s (1 - \xi)^{\#B - s}}{\prod_{x \in B} [1 - \xi + \xi m(x)]} F(d\xi) \tag{2.3}$$

(for finite $B \subseteq E$ and $0 \le s \le \#B$) verify (2.2). Hence, the corresponding random selection process indeed exists. In this case, the random variables $\{N(x)\}_{x \in E}$ are conditionally independent given $\xi \in [0, 1]$, with

$$\mathrm{P}\big\{N(x) = 1 \mid \xi\big\} = \frac{\xi m(x)}{1 - \xi + \xi m(x)}. \tag{2.4}$$

In particular, if F is concentrated at some $\xi \in [0, 1]$, then the random variables $\{N(x)\}_{x \in E}$ are independent. In either case, however, the $N(x)$ are neither identically distributed nor exchangeable (except for the case of a constant weight function; see Proposition 1.1).

Our purpose in this paper is to impose conditions on the weight function m of a random selection process $\boldsymbol{N}$ on (E, m) ensuring that its marginal distributions are necessarily as in (2.3) and, hence, the corresponding $\{N(x)\}_{x \in E}$ verify a de Finetti-type theorem (i.e., they are conditionally independent given ξ). Before stating our main results, we introduce some notation.

Given a denumerable set $E^\star := \{x_j^\star\}_{j\geq 1} \subseteq E$, we define $E_n^\star := \{x_1^\star, \ldots, x_n^\star\}$ for $n \geq 1$, and let

$$H(n) := \sum_{j=1}^{n} m\big(x_j^\star\big).$$

Also, we let $\hat{N}(x) := 1 - N(x)$, for $x \in E$. Therefore, $\hat{N}(B) := \sum_{x\in B} \hat{N}(x) = \#B - N(B)$, is the number of rejected elements in any finite set $B \subset E$. Finally, we define

$$\hat{H}(n) := \sum_{j=1}^{n} 1/m\big(x_j^\star\big).$$

Next, we state our main results.

Theorem 2.1 *Let $\boldsymbol{N}$ be a random selection process on (E, m). If there exists a denumerable set $E^\star = \{x_j^\star\}_{j\geq 1} \subseteq E$ such that $\{m(x_j^\star)\}_{j\geq 1}$ is monotone nonincreasing and $\sum_{j=1}^{\infty} m(x_j^\star) = \infty$, then*

(i) *The sequence*

$$\lambda_n := \frac{N(E_n^\star)}{H(n)}$$

converges almost surely as $n \to \infty$ to a random variable λ taking values in $[0, 1/\tilde{m}]$, where $\tilde{m} := \lim_j m(x_j^\star) \in [0, \infty)$.

(ii) *The random variables $\{N(x)\}_{x\in E}$ are conditionally independent given λ, with*

$$\mathrm{P}\big\{N(x) = 1 \mid \lambda\big\} = \frac{\lambda m(x)}{1 - \lambda\tilde{m} + \lambda m(x)}. \tag{2.5}$$

In particular, (2.3) and (2.4) hold with $\xi := \frac{\lambda}{1-\lambda\tilde{m}+\lambda} \in [0, 1]$, and $\xi = 1$ when $\tilde{m} = 0$ and $\lambda = \infty$.

The proof of Theorem 2.1 is given in Sect. 4. The strong law of large numbers in Theorem 2.1(i) establishes that the growth rate of $N(E_n^\star)$, the number of selected elements in $E_n^\star$, is $m(E_n^\star)$, except for the trivial cases $\lambda = 0$ and $\lambda = \infty$, in which all the elements of $x \in E$ are rejected and selected, respectively. An interesting feature of the de Finetti-type result in Theorem 2.1(ii) is that the existence of a denumerable set in E satisfying the corresponding hypotheses suffices to deduce the conditional independence and the expression of the marginal distributions of the *whole* random selection process.

The dual result of Theorem 2.1 for the nondecreasing case is the following.

Theorem 2.2 *Let $\boldsymbol{N}$ be a random selection process on (E, m). If there exists a denumerable set $E^\star = \{x_j^\star\}_{j\geq 1} \subseteq E$ such that $\{m(x_j^\star)\}_{j\geq 1}$ is monotone nondecreasing and $\sum_{j=1}^{\infty} 1/m(x_j^\star) = \infty$, then*

(i) *The random sequence*

$$\hat{\lambda}_n := \frac{\hat{N}(E_n^\star)}{\hat{H}(n)}$$

converges almost surely as $n \to \infty$ *to a random variable* $\hat{\lambda}$ *taking values in* $[0, \tilde{m}]$, *where* $\tilde{m} := \lim_j m(x_j^\star) \in (0, \infty]$.

(ii) *The random variables* $\{N(x)\}_{x \in E}$ *are conditionally independent given* $\hat{\lambda}$ *with*

$$\mathrm{P}\big\{N(x) = 1 \mid \hat{\lambda}\big\} = \frac{(1 - \hat{\lambda}/\tilde{m})m(x)}{\hat{\lambda} + (1 - \hat{\lambda}/\tilde{m})m(x)}. \tag{2.6}$$

In this case, (2.3) *and* (2.4) *hold with* $\xi := \frac{1-\hat{\lambda}/\tilde{m}}{1-\hat{\lambda}/\tilde{m}+\hat{\lambda}} \in [0, 1]$, *and* $\xi = 0$ *when* $\tilde{m} = \infty$ *and* $\hat{\lambda} = \infty$.

Hence, as a consequence of the strong law of large numbers in Theorem 2.2(i), the growth rate of the rejected elements in $E_n^\star$ is $\sum_{j=1}^n 1/m(x_j^\star)$, except for the cases $\hat{\lambda} = 0$ and $\hat{\lambda} = \infty$. The proof of Theorem 2.2 is also given in Sect. 4.

3 An example

Now we propose an example of a random selection process $\boldsymbol{N}$ with non constant weight function m (that is, the indicator random variables $N(x)$ are not exchangeable; recall Proposition 1.1).

Suppose that a basin is filled at a randomly chosen and unobservable constant flow of z particles per time unit. We assume that filling takes place at times labeled $t = 1, 2, \ldots$. Our goal is to estimate the flow z. To this end, we put into the basin some "special" and distinguishable particles. More precisely, we suppose that, at time $t \geq 1$, we put $\Delta_t \geq 0$ particles into the basin, so that, at time $t \geq 1$, there are zt particles coming from the filling, and

$$y(t) = \Delta_1 + \cdots + \Delta_t$$

special particles.

At each time $t \geq 1$ we draw (with replacement) a particle from the basin, and, upon this sequence of observations, we must estimate the flow z. For each $t = 1, 2, \ldots$, let $N(t) := 1$ if the particle drawn at time t is among the $y(t)$ special ones that we threw into the basin, and let $N(t) := 0$ otherwise. (Let us mention that, if z is assumed to have a continuous distribution, then we can think of the special particles as volumes added to the basin, and the results below remain valid.)

We can verify that Conditions **C1** and **C2** in the definition of a random selection process hold. As an illustration, let us check Condition **C2**. Let B be a finite set of time labels and fix $t \in B$. Then, letting F be the distribution of z,

$$\mathrm{P}\big\{N(t) = 1 \mid N(B) = 1\big\} = \frac{\int \frac{y(t)}{zt} \prod_{x \in B} \frac{zx}{y(x)+zx} \, F(dz)}{\sum_{w \in B} \int \frac{y(w)}{zw} \prod_{x \in B} \frac{zx}{y(x)+zx} \, F(dz)} = \frac{y(t)/t}{\sum_{w \in B} y(w)/w},$$

so that **C2** holds for the weight function $m(t) := y(t)/t$ for $t = 1, 2, \ldots$. Condition **C1** is verified similarly.

Consequently, the random variables $\{N(t)\}_{t\geq 1}$ are a random selection process with weight function

$$m(t) = \frac{y(t)}{t} = \frac{\Delta_1 + \cdots + \Delta_t}{t},$$

for $t \geq 1$. Hence, they are not exchangeable, except for the case of a constant Δ_t.

Regarding the estimation of the flow z, there are some cases of special interest. Suppose that $y(t)$, for $t \geq 1$, is of the form at^{α}, where $a > 0$ and $\alpha \geq 0$ are given constants, so that the weight function is $m(t) = at^{\alpha-1}$, for $t \geq 1$. Then we distinguish the following alternatives (in which the case $\alpha = 1$, corresponding to exchangeability, is discarded):

Case $0 \leq \alpha < 1$. The weight function is $m(t) = at^{\alpha-1}$, for $t \geq 1$, so that a de Finetti theorem holds and, in addition, by the strong law of large numbers (see Theorem 2.1),

$$\frac{a\sum_{t=1}^{n} t^{\alpha-1}}{\sum_{t=1}^{n} N(t)}$$

is an almost surely consistent estimator of z as $n \to \infty$. In particular, if $\alpha = 0$ then we put a particles into the basin at time $t = 1$, while $\Delta_2 = \Delta_3 = \cdots = 0$. In this case,

$$\lim_{n\to\infty} \frac{a \log n}{\sum_{t=1}^{n} N(t)} = z$$

with probability one.

Case $1 < \alpha \leq 2$. Now, by Theorem 2.2, the de Finetti theorem still holds and

$$\lim_{n\to\infty} \frac{a(n - \sum_{t=1}^{n} N(t))}{\sum_{t=1}^{n} t^{1-\alpha}} = z$$

with probability one. If $\alpha = 2$, then

$$\lim_{n\to\infty} \frac{a(n - \sum_{t=1}^{n} N(t))}{\log n} = z$$

almost surely.

Case $\alpha > 2$. The weights $m(t)$ are such that $m(t) \uparrow \infty$ as $t \to \infty$, though

$$\sum_{t=1}^{\infty} 1/m(t) < \infty.$$

In this case, even if the random variables $N(t)$ are conditionally independent given z, we do not obtain a law of large numbers because $\sum_t (1 - N(t))$ is finite with probability one (that is, we draw a finite number of particles coming from the filling at flow z). The interpretation is that, for $\alpha > 2$, we put so many special particles into the basin that the ones whose flow is to be estimated become asymptotically negligible.

4 Proofs

We begin this section with the proof of the equivalence between the exchangeability of the random indicator functions $\{N(x)\}_{x\in E}$ and the fact that the weight function m is constant.

Proof of Proposition 1.1 First, suppose that the random variables $\{N(x)\}_{x\in E}$ are exchangeable. Given $A=\{x_1,\dots,x_n\}$, we have

$$\mathrm{P}\big\{N(A)=1\big\}=n\cdot\mathrm{P}\big\{N(x_j)=1,\ N(A)=1\big\}\quad\forall j\in\{1,\dots,n\},$$

and so Condition **C2** holds for the constant weight function $m\equiv 1$. On the other hand, note that, by exchangeability,

$$\mathrm{P}\big\{N(x_1)=i_1,\dots,N(x_n)=i_n\big\}=:p_n(r)$$

depends only on $r:=i_1+\cdots+i_n$ and n (and it neither depends on A nor on the particular values of $i_1,\dots,i_n$). Now, given disjoint sets $A=\{x_1,\dots,x_n\}$ and $B=\{x_1',\dots,x_m'\}$,

$$\mathrm{P}\big\{N(x_k)=i_k,\ k=1,\dots,n,N(x_l')=i_l',\ l=1,\dots,m\mid N(A)=r\big\}=\frac{p_{n+m}(r+s)}{\mathrm{P}\{N(A)=r\}},$$

where $r=\sum_{k=1}^n i_k$ and $s=\sum_{l=1}^m i_l'$. Similarly,

$$\mathrm{P}\big\{N(x_k)=i_k,\ k=1,\dots,n\mid N(A)=r\big\}=\frac{1}{\binom{n}{r}},$$

and

$$\mathrm{P}\big\{N\big(x_l'\big)=i_l',\ l=1,\dots,m\mid N(A)=r\big\}=\frac{p_{n+m}(r+s)\binom{n}{r}}{\mathrm{P}\{N(A)=r\}}.$$

It follows that Condition **C1** holds.

Conversely, let us assume that the indicator random variables $\{N(x)\}_{x\in E}$ satisfy Conditions **C1** and **C2** for the constant weight function $m\equiv 1$. To prove the exchangeability property, it suffices to show that the distribution of $\boldsymbol{N}$ is invariant by transpositions. To this end, given $A=\{x_1,\dots,x_n\}$ we will show that

$$\mathrm{P}\big\{N(x_1)=i_1,\ N(x_2)=i_2,\ N(x_3)=i_3,\dots,N(x_n)=i_n\big\} \tag{4.1}$$

$$=\mathrm{P}\big\{N(x_1)=i_2,\ N(x_2)=i_1,\ N(x_3)=i_3,\dots,N(x_n)=i_n\big\}, \tag{4.2}$$

where, in order to avoid trivial cases, we will suppose that $i_1+i_2=1$. Observe that (4.1) equals

$$\mathrm{P}\big\{N(x_1)=i_1,N(x_2)=i_2\mid N\{x_1,x_2\}=1,\ N(x_k)=i_k,\ 3\le k\le n\big\}$$
$$\times\,\mathrm{P}\big\{N(x_k)=i_k,\ 3\le k\le n\big\},$$

which, by Conditions **C1** and **C2**, equals

$$\frac{1}{2} \cdot \mathrm{P}\big\{N(x_k) = i_k,\ 3 \le k \le n\big\}.$$

Proceeding similarly with (4.2), the equality in (4.1)–(4.2) easily follows. □

In what follows, we consider an infinite set E and a weight function $m : E \to (0, \infty)$. Let $E^\star = \{x_j^\star\}_{j\ge1} \subseteq E$ be an arbitrary denumerable set, and let $E_n^\star := \{x_1^\star, \dots, x_n^\star\}$ for $n \ge 1$. For ease of notation, we will write $m_j := m(x_j^\star)$, for $j \ge 1$. Recall also that

$$H(n) := \sum_{j=1}^{n} m_j, \qquad \hat{H}(n) := \sum_{j=1}^{n} 1/m_j, \quad \text{and} \quad \tilde{m} := \lim_j m_j, \tag{4.3}$$

provided that the limit exists.

The following lemma analyzes the asymptotic behavior, as $n \to \infty$, of

$$\frac{M_{r_n - s}(E_n^\star)}{M_{r_n}(E_n^\star)},$$

where $s \ge 1$ is a fixed integer and $s \le r_n \le n$, for $n \ge 1$. In the forthcoming, to simplify the notation, we will drop the index n and the sequence $\{r_n\}$ will be denoted by r.

Lemma 4.1 *Suppose that the sequence of weights $\{m_j\}_{j\ge1}$ in $E^\star$ is nonincreasing and that $\sum_{j=1}^{\infty} m_j = \infty$. For fixed $s \ge 1$ and $s \le r \le n$ varying with n, we have*

$$\lim_{n\to\infty} \frac{M_{r-s}(E_n^\star)}{M_r(E_n^\star)} = \begin{cases} y^s & \text{if } \tilde{m} = 0 \text{ and } r/H(n) \to y \in [0, \infty], \\ \frac{y^s}{\tilde{m}^s(1-y)^s} & \text{if } \tilde{m} > 0 \text{ and } r/n \to y \in [0, 1]. \end{cases}$$

Proof First of all, observe that

$$\sum_{1\le j_1<\cdots<j_r\le n} m_{j_1}\cdots m_{j_r}\left[H(n) - \sum_{i=1}^{r} m_{j_i}\right] = (r+1)M_{r+1}\big(E_n^\star\big)$$

and, since $\{m_j\}$ is nonincreasing, we have

$$M_r\big(E_n^\star\big)\big[H(n) - H(r)\big] \le (r+1)M_{r+1}\big(E_n^\star\big) \le M_r\big(E_n^\star\big)H(n-r),$$

or

$$\frac{r+1}{H(n)}\frac{H(n)}{H(n-r)} \le \frac{M_r(E_n^\star)}{M_{r+1}(E_n^\star)} \le \frac{r+1}{H(n)}\frac{H(n)}{H(n)-H(r)}.$$

Consider first the case $\tilde{m} = 0$. If $r/H(n) \to y \in (0, \infty)$, then

$$\frac{H(n)}{H(n)-H(r)} \to 1 \quad \text{and} \quad 1 \le \frac{H(n)}{H(n-r)} \le \frac{H(n)}{H(n)-H(r)},$$

and thus

$$\frac{M_r(E_n^\star)}{M_{r+1}(E_n^\star)} \to y \quad \text{and} \quad \frac{M_{r-s}(E_n^\star)}{M_r(E_n^\star)} \to y^s,$$

which proves the stated result. The above limit also holds when $r/H(n) \to y = \infty$ (indeed, $H(n)/H(n-r) \geq 1$), and also when $y = 0$ because $(r+1)/[H(n) - H(r)] \to 0$ (regardless r is bounded or not). This completes the proof for the case $\tilde{m} = 0$.

Now we address the case $\tilde{m} > 0$. Assume first that $r/n \to y \in (0, 1)$. Since

$$\frac{H(n)}{H(n) - H(r)} \sim \frac{n}{n-r} \quad \text{and} \quad \frac{H(n)}{H(n-r)} \sim \frac{n}{n-r},$$

we have

$$\frac{M_r(E_n^\star)}{M_{r+1}(E_n^\star)} \to \frac{y}{\tilde{m}(1-y)} \quad \text{and} \quad \frac{M_{r-s}(E_n^\star)}{M_r(E_n^\star)} \to \frac{y^s}{\tilde{m}^s(1-y)^s},$$

as we wanted to prove. The stated result also holds for $y = 0$ because $(r+1)/H(n) \to 0$ and $H(n)/[H(n) - H(r)] \to 1$, and it is also true for $y = 1$ because $H(n)/H(n-r) \to \infty$. □

Next, we state a dual result for a denumerable subset $E^\star \subseteq E$ with nondecreasing weights $\{m_j\}_{j\geq 1}$.

Lemma 4.2 *Suppose that the sequence of weights $\{m_j\}_{j\geq 1}$ in $E^\star$ is nondecreasing and that $\sum_{j=1}^\infty 1/m_j = \infty$. Fix $s \geq 1$ and suppose that $s \leq r \leq n$ varies with n. Then*

$$\lim_{n\to\infty} \frac{M_{n-r+s}(E_n^\star)}{M_{n-r}(E_n^\star)} = \begin{cases} y^s & \text{if } \tilde{m} = \infty \text{ and } r/\hat{H}(n) \to y \in [0, \infty], \\ \frac{\tilde{m}^s y^s}{(1-y)^s} & \text{if } \tilde{m} < \infty \text{ and } r/n \to y \in [0, 1]. \end{cases}$$

Proof Note that

$$\frac{M_{n-r+s}(E_n^\star)}{M_{n-r}(E_n^\star)} = \frac{\hat{M}_{r-s}(E_n^\star)}{\hat{M}_r(E_n^\star)}$$

where

$$\hat{M}_r\big(E_n^\star\big) := \frac{M_{n-r}(E_n^\star)}{\prod_{j=1}^n m_j} = \sum_{1\leq j_1<\cdots<j_r\leq n} \frac{1}{m_{j_1}\cdots m_{j_r}},$$

which is precisely the "binomial" coefficient $M_r(E_n)$ corresponding to the weight sequence $\{\hat{m}_j = 1/m_j\}_{j\geq 1}$. Now, the stated result directly follows from Lemma 4.1. □

For the so-called nonincreasing case, we obtain the next result, which uses the notation introduced in (4.3).

Proposition 4.3 *Assume that E contains a denumerable set $E^\star = \{x_j^\star\}_{j\geq 1}$ such that $\{m(x_j^\star)\}$ is nonincreasing and $\sum_{j=1}^\infty m(x_j^\star) = \infty$. Consider an arbitrary finite set $B \subseteq E - E^\star$ and any $0 \leq s \leq \#B$.*

(i) *When $\tilde{m} = 0$, if r varies with n in such a way that $0 \leq r \leq n + \#B$ and $r/H(n) \to y \in [0, \infty)$, then*

$$\lim_{n\to\infty} \frac{M_{r-s}(E_n^\star)}{M_r(B \cup E_n^\star)} = \frac{y^s}{\prod_{x\in B}[1 + ym(x)]}. \tag{4.4}$$

If $y = \infty$, the limit is 0 for $s < \#B$, and it equals $1/M_{\#B}(B)$ for $s = \#B$.

(ii) *When $\tilde{m} > 0$, if r varies with n in such a way that $0 \leq r \leq n + \#B$ and $r/n \to y \in [0, 1]$, then*

$$\lim_{n\to\infty} \frac{M_{r-s}(E_n^\star)}{M_r(B \cup E_n^\star)} = \frac{y^s(1-y)^{\#B-s}\tilde{m}^{\#B-s}}{\prod_{x\in B}[(1-y)\tilde{m} + ym(x)]}. \tag{4.5}$$

Proof If $\tilde{m} = 0$ and $y > 0$, or $\tilde{m} > 0$ and $0 < y < 1$, then the statements (4.4) and (4.5), respectively, follow from Lemma 4.1 and the identity

$$\frac{M_r(B \cup E_n^\star)}{M_{r-s}(E_n^\star)} = \sum_{t=0\vee(r-n)}^{\#B\wedge r} M_t(B)\frac{M_{r-t}(E_n^\star)}{M_{r-s}(E_n^\star)} \tag{4.6}$$

(indeed, in either case, $\#B < r < n$ for n large enough).

If $y = 0$, then $r < n$ for large n, and the term $t = 0$ in (4.6) converges to ∞ if $s > 0$; while, for $s = 0$, the first term is 1 and the remaining terms converge to 0. The results for $y = \infty$ when $\tilde{m} = 0$, or $y = 1$ when $\tilde{m} > 0$, are easily analyzed in the same way (in this case, the relevant term of (4.6) is $t = \#B$). □

The dual result for nondecreasing sequences is the following.

Proposition 4.4 *Assume that E contains a denumerable set $E^\star = \{x_j^\star\}_{j\geq 1}$ such that $\{m(x_j^\star)\}$ is nondecreasing and $\sum_{j=1}^\infty 1/m(x_j^\star) = \infty$. Consider an arbitrary finite set $B \subseteq E - E^\star$ and any $0 \leq s \leq \#B$.*

(i) *When $\tilde{m} = \infty$, if r varies with n in such a way that $0 \leq r \leq n + \#B$ and $r/\hat{H}(n) \to y \in [0, \infty)$, then*

$$\lim_{n\to\infty} \frac{M_{n+\#B-r-s}(E_n^\star)}{M_{n+\#B-r}(B \cup E_n^\star)} = \frac{y^{\#B-s}}{\prod_{x\in B}[y + m(x)]}.$$

If $y = \infty$, then the limit is 0 for $s > 0$, and it equals 1 for $s = 0$.

(ii) *When $\tilde{m} < \infty$, if r varies with n in such a way that $0 \leq r \leq n + \#B$ and $r/n \to y \in [0, 1]$, then*

$$\lim_{n\to\infty} \frac{M_{n+\#B-r-s}(E_n^\star)}{M_{n+\#B-r}(B \cup E_n^\star)} = \frac{(1-y)^s y^{\#B-s}\tilde{m}^{\#B-s}}{\prod_{x\in B}[y\tilde{m} + (1-y)m(x)]}.$$

Proof The proof is similar to that of Proposition 4.3. Indeed, just replace (4.6) with

$$\frac{M_{n+\#B-r}(B\cup E_n^\star)}{M_{n+\#B-r-s}(E_n^\star)}=\sum_{t=0\vee(\#B-r)}^{\#B\wedge(n+\#B-r)}M_t(B)\frac{M_{n+\#B-r-t}(E_n^\star)}{M_{n+\#B-r-s}(E_n^\star)}$$

and proceed similarly. □

We will also need the following result.

Lemma 4.5 *Let* $\boldsymbol{N}$ *be a random selection process on* (E,m), *and let* $E^\star=\{x_j^\star\}_{j\geq 1}$ *be any denumerable set in* E. *For any finite* $B\subseteq E-E^\star$, *consider the reverse filtration*

$$\begin{aligned}\mathcal{G}_n&:=\sigma\big(N\big(B\cup E_n^\star\big),N\big(B\cup E_{n+1}^\star\big),\ldots\big)\\&=\sigma\big(N\big(B\cup E_n^\star\big),N\big(x_{n+1}^\star\big),N\big(x_{n+2}^\star\big),\ldots\big),\quad \textit{for } n\geq 1.\end{aligned}$$

Then, for any $0\leq s\leq \#B$,

$$Z_n(B,s):=\mathrm{P}\big\{N(B)=s\mid\mathcal{G}_n\big\}=\left[\frac{M_s(B)M_{r-s}(E_n^\star)}{M_r(B\cup E_n^\star)}\right]_{r=N(B\cup E_n^\star)}\tag{4.7}$$

is a reverse martingale that converges almost surely to $Z_\infty(B,s):=\mathrm{P}\{N(B)=s\mid\mathcal{G}_\infty\}$, *where* $\mathcal{G}_\infty:=\bigcap_n\mathcal{G}_n$.

Proof First of all, note that the rightmost equality in (4.7) is a direct consequence of the conditional independence in **C1** and (2.1). To proceed with the proof, it is clear that $Z_n(B,s)=\mathrm{P}\{N(B)=s\mid\mathcal{G}_n\}$ is a reverse martingale with respect to the reverse filtration $\{\mathcal{G}_n\}_{n\geq 1}$, and so $Z_n\xrightarrow{\text{a.s.}}Z_\infty=\mathrm{P}\{N(B)=s\mid\mathcal{G}_\infty\}$; see (Fristedt and Gray 1997, Theorem 24, p. 484). □

Observe that $\{Z_n(B,s)\}_{0\leq s\leq\#B}$ and $\{Z_\infty(B,s)\}_{0\leq s\leq\#B}$ are probability distributions (with probability one) and, furthermore,

$$\mathrm{E}\big[Z_n(B,s)\big]=\mathrm{E}\big[Z_\infty(B,s)\big]=\mathrm{P}\big\{N(B)=s\big\}.$$

Now we are ready to prove our main results in this paper.

Proof of Theorem 2.1. Proof of (i). We will first examine the case $\tilde{m}=0$. Fix any finite set $B\subseteq E-E^\star$ and let $\bar{\Omega}$ be the set of probability one where $Z_n(B,\#B)\to Z_\infty(B,\#B)$ (recall Lemma 4.5). Observe that, in $\bar{\Omega}$,

$$Z_\infty(B,\#B)=\lim_{n\to\infty}\left[\frac{M_{\#B}(B)M_{r-\#B}(E_n^\star)}{M_r(B\cup E_n^\star)}\right]_{r=N(B\cup E_n^\star)}.\tag{4.8}$$

We distinguish the following cases.

– if $0 < \liminf \lambda_n = \underline{\lambda} \le \bar{\lambda} = \limsup \lambda_n < \infty$, then there would exist subsequences n' and n'' with $N(B \cup E^\star_{n'})/H(n') \to \underline{\lambda}$ and $N(B \cup E^\star_{n''})/H(n'') \to \bar{\lambda}$, so that, according to Proposition 4.3(i), we would have

$$Z_\infty(B, \#B) = \frac{M_{\#B}(B)\underline{\lambda}^{\#B}}{\prod_{x\in B}[1+\underline{\lambda}m(x)]} = \frac{M_{\#B}(B)\bar{\lambda}^{\#B}}{\prod_{x\in B}[1+\bar{\lambda}m(x)]}.$$

Hence, $\underline{\lambda} = \bar{\lambda}$ because $\prod_{x\in B}[1/\lambda + m(x)]^{-1}$ is an increasing function of λ on $(0, \infty)$;

– if $\liminf \lambda_n = 0$ then, along a subsequence n' with $N(B \cup E^\star_{n'})/H(n') \to 0$, Proposition 4.3(i) yields $Z_\infty(B, \#B) = 0$ and, necessarily, $\lim \lambda_n = 0$;

– if $\limsup \lambda_n = \infty$ then, along a subsequence n' with $N(B \cup E^\star_{n'})/H(n') \to \infty$, we would have $Z_\infty(B, \#B) = 1$, and thus $\lim \lambda_n = \infty$.

In summary, $\lambda_n \xrightarrow{\text{a.s.}} \lambda \in [0, \infty]$, as we wanted to prove. Also note that

$$Z_\infty(B, \#B) = \begin{cases} \frac{M_{\#B}(B)\lambda^{\#B}}{\prod_{x\in B}[1+\lambda m(x)]} & \text{if } 0 \le \lambda < \infty, \\ 1 & \text{if } \lambda = \infty. \end{cases} \tag{4.9}$$

For the case $\tilde{m} > 0$, from (4.8) and since $H(n) \sim n\tilde{m}$, a similar argument gives $\lambda_n \to \lambda \in [0, 1/\tilde{m}]$ in $\bar{\Omega}$. This establishes the stated result. Alternatively, we have $N(E^\star_n)/n \longrightarrow \lambda\tilde{m} \in [0, 1]$. Also notice that, according to Proposition 4.3(ii),

$$Z_\infty(B, \#B) = \frac{M_{\#B}(B)\lambda^{\#B}}{\prod_{x\in B}[1-\lambda\tilde{m}+\lambda m(x)]}. \tag{4.10}$$

The proof of statement (i) is now complete. We proceed to prove the de Finetti-type result.

Proof of (ii). First of all, we consider the case $\tilde{m} > 0$. Since λ is $\mathcal{G}_\infty$-measurable (by Lemma 4.5) and $Z_\infty(B, \#B)$, given by (4.10), is a measurable function of λ, we have

$$\mathrm{P}\big\{N(B) = \#B \mid \lambda\big\} = \frac{M_{\#B}(B)\lambda^{\#B}}{\prod_{x\in B}[1-\lambda\tilde{m}+\lambda m(x)]}.$$

Moreover, for any $s < \#B$, since $N(B \cup E^\star_n)/n \to \lambda\tilde{m} \in [0, 1]$, Proposition 4.3(ii) easily gives

$$Z_\infty(B, s) = \frac{M_s(B)\lambda^s(1-\lambda\tilde{m})^{\#B-s}}{\prod_{x\in B}[1-\lambda\tilde{m}+\lambda m(x)]},$$

and hence

$$\mathrm{P}\big\{N(B) = s \mid \lambda\big\} = \frac{M_s(B)\lambda^s(1-\lambda\tilde{m})^{\#B-s}}{\prod_{x\in B}[1-\lambda\tilde{m}+\lambda m(x)]} = \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+\xi m(x)]}$$

(recall the definition of ξ given in Theorem 2.1). Hence, conditional on λ—or ξ—the random variables $N(x)$ are independent; thus, they verify (2.4) and (2.5).

Turning our attention to the case $\tilde{m} = 0$, from (4.9) we now deduce

$$\begin{aligned}\mathrm{P}\big\{N(B) = \#B \mid \lambda\big\} &= \frac{M_{\#B}(B)\lambda^{\#B}}{\prod_{x\in B}[1+\lambda m(x)]}\mathbb{1}_{\{\lambda<\infty\}} + \mathbb{1}_{\{\lambda=\infty\}} \\ &= \frac{M_{\#B}(B)\xi^{\#B}}{\prod_{x\in B}[1-\xi+\xi m(x)]},\end{aligned}$$

where $\mathbb{1}$ stands for the indicator function.

Moreover, for any $s < \#B$, by Proposition 4.3(i), we have

$$Z_\infty(B,s) = \frac{M_s(B)\lambda^s}{\prod_{x\in B}[1+\lambda m(x)]}\mathbb{1}_{\{\lambda<\infty\}} = \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+\xi m(x)]}.$$

This proves the conditional independence of the $N(x)$ given λ—or ξ—as well as (2.4) and (2.5). □

Recalling the notation $\hat{N}(A) = \#A - N(A)$, for $A \subset E$, note that (4.7) becomes

$$Z_n(B,s) = \left[\frac{M_s(B)M_{n+\#B-r-s}(E_n^\star)}{M_{n+\#B-r}(B\cup E_n^\star)}\right]_{r=\hat{N}(B\cup E_n^\star)}.$$

We will now use Proposition 4.4 to prove the de Finetti-type theorem when E contains a denumerable set with nondecreasing weights.

Proof of Theorem 2.2. Proof of (i). In case that $\tilde{m} = \infty$, for a fixed finite set $B \subseteq E - E^\star$ let $\bar{\Omega}$ be the set of probability one where $Z_n(B,0) \to Z_\infty(B,0)$ (see Lemma 4.5). Proceeding as in the proof of Theorem 2.1(i) and according to Proposition 4.4(i), we conclude that

$$\hat{\lambda}_n \xrightarrow{\text{a.s.}} \hat{\lambda} \in [0,\infty] \quad \text{and} \quad Z_\infty(B,0) = \begin{cases} \dfrac{\hat{\lambda}^{\#B}}{\prod_{x\in B}[\hat{\lambda}+m(x)]} & \text{if } 0 \le \hat{\lambda} < \infty, \\ 1 & \text{if } \hat{\lambda} = \infty. \end{cases} \tag{4.11}$$

When $\tilde{m} < \infty$, since $\hat{H}(n)/n \to 1/\tilde{m}$, we can similarly prove, according to Proposition 4.4(ii), that

$$\hat{\lambda}_n \xrightarrow{\text{a.s.}} \hat{\lambda} \in [0,\tilde{m}] \quad \text{and} \quad Z_\infty(B,0) = \frac{\hat{\lambda}^{\#B}}{\prod_{x\in B}[\hat{\lambda}+(1-\hat{\lambda}/\tilde{m})m(x)]}.$$

Proof of (ii). For $\tilde{m} < \infty$, since $\hat{N}(B\cup E_n^\star)/n \to \hat{\lambda}/\tilde{m} \in [0,1]$, Proposition 4.4(ii) shows that, for any $0 \le s \le \#B$,

$$Z_\infty(B,s) = \frac{M_s(B)(1-\hat{\lambda}/\tilde{m})^s\hat{\lambda}^{\#B-s}}{\prod_{x\in B}[\hat{\lambda}+(1-\hat{\lambda}/\tilde{m})m(x)]} = \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+\xi m(x)]},$$

which yields the expression of $\mathrm{P}\{N(B) = s \mid \hat{\lambda}\} = \mathrm{P}\{N(B) = s \mid \xi\}$. This proves the conditional independence of the random variables $\{N(x)\}_{x\in E}$ given $\hat{\lambda}$ or ξ, and (2.4) and (2.6) hold.

If $\tilde{m} = \infty$, then Proposition 4.4(i) gives, for $s > 0$,

$$Z_\infty(B, s) = \frac{M_s(B)\hat{\lambda}^{\#B-s}}{\prod_{x\in B}[\hat{\lambda} + m(x)]}\mathbb{1}_{\{\hat{\lambda}<\infty\}} = \frac{M_s(B)\xi^s(1-\xi)^{\#B-s}}{\prod_{x\in B}[1-\xi+\xi m(x)]}.$$

From (4.11), the above expression holds as well for $s = 0$. In the same way as in the preceding case, the conditional independence and (2.4) or (2.6) follow readily. □

5 The rejection function

It was pointed out in Lemmas 3.1, 3.2, and 3.4 in Vélez Ibarrola and Prieto-Rumeau (2009) that the marginal distributions of a random selection process $\boldsymbol{N}$ on (E, m) are always determined by the function

$$p_0(B) := \mathrm{P}\big\{N(B) = 0\big\}, \quad \text{for finite } B \subset E,$$

which is a symmetric function of the vector $[m(x)]_{x\in B}$. Further, if m is injective, then $p_0(B)$ is determined by the *rejection function* $p_0(\{x\})$, for $x \in E$, that gives the probability of rejecting each point x. Under the hypotheses of Theorems 2.1 or 2.2, we know that

$$p_0(B) = \int_0^1 \frac{(1-\xi)^{\#B}}{\prod_{x\in B}[1-\xi+\xi m(x)]}F(d\xi) \quad \text{and}$$

$$p_0(\{x\}) = \int_0^1 \frac{1-\xi}{1-\xi+\xi m(x)}F(d\xi)$$

for some distribution F on $[0, 1]$. However, we can look for further characterizations of the functions $p_0(B)$ or $p_0(\{x\})$.

Let $I := \{m(x) \mid x \in E\} \subseteq R_+$ be the weights range. We will deal with the case when I is an infinite set, and we will use the following operators:

$$\Delta_h g(m_1, \ldots, m_k) := \frac{g(m_1, \ldots, m_k + h) - g(m_1, \ldots, m_k)}{h},$$

$$\Lambda_h g(m_1, \ldots, m_k) := \frac{(m_k + h)g(m_1, \ldots, m_k + h) - m_k g(m_1, \ldots, m_k)}{h}.$$

A function $g_0 : I \cup \{0\} \to [0, 1]$ with $g_0(0) = 1$ will be said *totally decreasing on* I if

$$(-1)^k \Delta_{m_k-m_{k-1}}\Delta_{m_{k-1}-m_{k-2}}\cdots\Delta_{m_3-m_2}\Delta_{m_2-m_1}g_0(m_1) \geq 0 \tag{5.1}$$

and, in addition,

$$\Lambda_{m_k-m_{k-1}}\Lambda_{m_{k-1}-m_{k-2}}\cdots\Lambda_{m_3-m_2}\Lambda_{m_2-m_1}g_0(m_1) \geq 0 \tag{5.2}$$

for all different values $m_1 \in I \cup \{0\}$ and $m_2, \ldots, m_k \in I$.

Let us define, for all $m \in I$,

$$g_1(m) := -\Delta_m g_0(0) = \frac{1 - g_0(m)}{m}.$$

Next, for all $m_1 \neq m_2 \in I$,

$$g_0(m_1, m_2) := \Lambda_{m_2 - m_1} g_0(m_1) = \frac{m_2 g_0(m_2) - m_1 g_0(m_1)}{m_2 - m_1},$$

$$g_1(m_1, m_2) := -\Delta_{m_2 - m_1} g_0(m_1) = \frac{g_0(m_1) - g_0(m_2)}{m_2 - m_1},$$

$$g_2(m_1, m_2) := \Delta_{m_2 - m_1} \Delta_{m_1} g_0(0) = 1 - \frac{m_2 g_0(m_1) - m_1 g_0(m_2)}{m_2 - m_1}.$$

Recursively, for all finite subsets $J = \{m_1, m_2, \ldots, m_n\} \subseteq I$ and $0 < k \leq n$,

$$g_0(J) := g_0(m_1, m_2, \ldots, m_n) = \Lambda_{m_n - m_{n-1}} g_0(m_1, m_2, \ldots, m_{n-1}),$$
$$g_k(J) := g_k(m_1, m_2, \ldots, m_n) = -\Delta_{m_n - m_{n-1}} g_{k-1}(m_1, m_2, \ldots, m_{n-1}).$$

Thus, it is easy to prove the next result.

Lemma 5.1 *If g_0 is totally decreasing on I, then the $g_k(J)$ are symmetric nonnegative functions such that $\{q_k(J) := M_k(J) g_k(J)\}_{0 \leq k \leq \#J}$ is a probability distribution on $\{0, 1, \ldots, \#J\}$. Furthermore, these probability distributions are related by means of the equivalent formulas*

$$q_k(J) = \sum_{r=0}^{\#K} q_{k+r}(J \cup K) \frac{M_k(J) M_r(K)}{M_{k+r}(J \cup K)} \quad \text{and} \quad g_k(J) = \sum_{r=0}^{\#K} g_{k+r}(J \cup K) M_r(K).$$

(Clearly, (5.1) and (5.2) are used to ensure that $g_k(J) \geq 0$.) This lemma states that any totally decreasing function g_0 on I defines (the marginal distributions of) a random selection process on (I, ν) where ν is the identity function: $\nu(m) = m$. According to Theorems 2.1 and 2.2 we conclude the following.

Theorem 5.2 *Let $I \subseteq R_+$ contain either a decreasing sequence $\{m_j\}$ with $\sum_{j=1}^{\infty} m_j = \infty$ or an increasing sequence $\{m_j\}$ with $\sum_{j=1}^{\infty} 1/m_j = \infty$. Then g_0 is a totally decreasing function on I if and only if there exists a probability distribution F on $[0, 1]$ such that*

$$g_0(m) = \int_0^1 \frac{1 - \xi}{1 - \xi + \xi m} F(d\xi) \tag{5.3}$$

for all $m \in I$. More generally, for all finite $J \subseteq I$ and $0 \leq k \leq \#J$,

$$g_k(J) = \int_0^1 \frac{\xi^k (1 - \xi)^{\#J - k}}{\prod_{m \in J} [1 - \xi + \xi m]} F(d\xi).$$

Thus, if (E, m) is such that $I = \{m(x) \mid x \in E\}$ satisfies the hypothesis of Theorem 5.2, then $p_0(\{x\})$ is the *rejection function* of a random selection process on (E, m) if and only if

$$p_0\big(\{x\}\big) = g_0\big(m(x)\big)$$

for some *totally decreasing* function g_0 on I. Nevertheless, what seems more interesting is that Theorem 5.2 gives the integral representation (5.3) for totally decreasing functions, while those functions were defined, as indicated in (5.1) and (5.2), in terms of the increments Δ_h and Λ_h, successively applied with different spans h.

Obviously, a function $g_0(m)$ on I satisfying (5.3) can be extended to $[0, \infty)$ as

$$g_0(m) := \int_0^\infty \frac{1}{1+\rho m} G(d\rho) = \int_0^\infty e^{-mx} f(x)\, dx,$$

where G is the (possibly defective) distribution of $\rho := \xi/(1-\xi)$, and

$$f(x) := \int_0^\infty \frac{1}{\rho} e^{-x/\rho} G(d\rho).$$

Therefore, $g_0(m)$ is a completely monotone function, and it is the Laplace transform of a probability density $f(x)$ (since $g_0(0) = 1$), which is also completely monotone (see Feller 1966, Chap. XIII, §4).

Finally, let us point out that Theorem 5.2 is, in this context, an alternative tool to the Hausdorff moment problem result. In fact, if (E, m) satisfies the hypothesis of Theorem 2.1 or 2.2, but I does not verify the assumption of Theorem 5.2, then E necessarily contains a denumerable set $E^\star = \{x_j^\star\}_{j\geq 1}$ with $m(x_j^\star) = m$ for some $m \in I$. In this case, it is no longer true that $p_0(\{x\})$ determines $p_0(B)$; but, it is easily seen from (2.2) that the sequence $c_n = p_0(E_n^\star)$ must be completely monotone:

$$(-1)^k \Delta^k c_n \geq 0 \quad \text{or} \quad p_0\big(E_n^\star\big) = \int_0^1 x^n G(dx) = \int_0^1 \frac{(1-\xi)^n}{(1-\xi+\xi m)^n} F(d\xi)$$

for some distribution function F on $[0, 1]$. Thus, the totally decreasing property of $g_0(m)$ is now replaced with the complete monotonicity of $p_0(E_n^\star)$.

6 Final comments

It is worthwhile to observe that when E is not countable, then the hypotheses of Theorems 2.1 or 2.2 are satisfied by *any weight function*. Indeed, at least one of the sets

$$\big\{x \in E \mid 1/(k+1) < m(x) \leq 1/k\big\} \quad \text{or} \quad \big\{x \in E \mid k < m(x) \leq k+1\big\}$$

is infinite for some integer k and, therefore, it contains some denumerable set $E^\star = \{x_j^\star\}_{j\geq 1}$ with monotone weights $m(x_j^\star)$ that are bounded. Thus, $\tilde{m} \in (0, \infty)$ and $\sum_{j=1}^\infty m(x_j^\star) = \infty$, as well as $\sum_{j=1}^\infty m(x_j^\star)^{-1} = \infty$.

On the contrary, when E is countable, there remain some open problems related to the family of marginal distributions in case that the weights sequence (or its inverse) is a convergent series.

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